09_year_prompt_effect

```r
source("R/00_setup.R")
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
c2 <- readRDS("c2_matched.rds")
mc <- readRDS("model_coords.rds")
gt <- readRDS("country_wave_coords.rds") %>% transmute(country = iso, wave, gt1 = Dim1, gt2 =

latest    <- latest_wave(c2)
latest_gt <- gt %>% group_by(country) %>% filter(wave == max(wave)) %>% ungroup()

c2d <- latest %>% transmute(model, country, dist_C2 = dist)

c1d <- mc %>% filter(condition == "C1") %>%
  group_by(model, country) %>%
  summarise(Dim1 = mean(Dim1), Dim2 = mean(Dim2), .groups = "drop") %>%
  inner_join(latest_gt, by = "country") %>%
  transmute(model, country, dist_C1 = sqrt((Dim1 - gt1)^2 + (Dim2 - gt2)^2))

pair <- inner_join(c1d, c2d, by = c("model", "country")) %>%
  mutate(improve = dist_C1 - dist_C2)
```

```r
#' ## Paired country bootstrap

bm <- boot_paired(pair, "improve")
report_ci(bm, "C1 distance minus C2 distance (> 0 = naming the year improves placement)")
```

```
=== C1 distance minus C2 distance (> 0 = naming the year improves placement) ===
                   X2.5.   X50. X97.5. vs_ref
Qwen/Qwen3.6-Plus -0.004  0.006  0.017   n.s.
claude-opus-4-8   -0.016 -0.005  0.006   n.s.
gemini-3.6-flash  -0.011  0.001  0.014   n.s.
gpt-5.5           -0.001  0.009  0.019   n.s.
```

```r
#' ## Wilcoxon signed-rank, per model

cat("\n=== Wilcoxon signed-rank (paired, per model), C1 vs C2 distance ===\n")
```

```
=== Wilcoxon signed-rank (paired, per model), C1 vs C2 distance ===
```

```r
for (m in unique(pair$model)) {
  d <- pair %>% filter(model == m)
  w <- wilcox.test(d$dist_C1, d$dist_C2, paired = TRUE)
  cat(sprintf("  %-22s median improve = %+.3f   p = %.3f\n", m, median(d$improve), w$p.value
}
```

```
  Qwen/Qwen3.6-Plus      median improve = +0.014   p = 0.271
  claude-opus-4-8        median improve = -0.013   p = 0.314
  gemini-3.6-flash       median improve = -0.004   p = 0.675
  gpt-5.5                median improve = +0.016   p = 0.026
```

```r
#' ## Scale of the effect
#'
#' Any gain from the year should be read against the two quantities around it:
#' the distance still remaining between the model and the truth, and the distance
#' naming the *country* moves the answer. A year effect an order of magnitude
#' smaller than the residual error is not a correction.

reloc <- readRDS("c0_relocation.rds")$reloc %>%
```

```r
  group_by(model) %>% summarise(mean_reloc = mean(reloc_dist), .groups = "drop")

cat("\n=== the year effect in context ===\n")
```

```
=== the year effect in context ===
```

```r
pair %>% group_by(model) %>%
  summarise(median_year_gain = median(improve), mean_remaining_dist = mean(dist_C2),
            .groups = "drop") %>%
  left_join(reloc, by = "model") %>%
  mutate(remaining_vs_gain = round(mean_remaining_dist / abs(median_year_gain), 1),
         country_move_vs_gain = round(mean_reloc / abs(median_year_gain), 1),
         across(where(is.numeric), ~ round(.x, 3))) %>% print()
```

```
# A tibble: 4 x 6
  model        median_year_gain mean_remaining_dist mean_reloc remaining_vs_gain
  <chr>                   <dbl>               <dbl>      <dbl>             <dbl>
1 Qwen/Qwen3.~            0.014               0.231      0.453              16.9
2 claude-opus~           -0.013               0.182      0.39               14.4
3 gemini-3.6-~           -0.004               0.151      0.545              35.3
4 gpt-5.5                 0.016               0.184      0.486              11.5
# i 1 more variable: country_move_vs_gain <dbl>
```

```r
saveRDS(pair, "year_prompt_effect.rds")
```